

Understanding similarity

Method	Result interpretation
Cosine similarity	Higher score = more similar
Dot product	Higher score = stronger similarity
Euclidean distance	Lower distance = more similar

1. Cosine Similarity

What it does: Measures the angle between vectors. Ignores vector length completely.

QA Example:

Test Case A: "User login with valid email and password"

Test Case B: "Valid user authentication using email credentials"

→ High cosine similarity (focuses on meaning, not length)

✓ Best for: Finding similar test cases regardless of description length

2. Dot Product

What it does: Like cosine similarity, but considers vector length too.

QA Example:

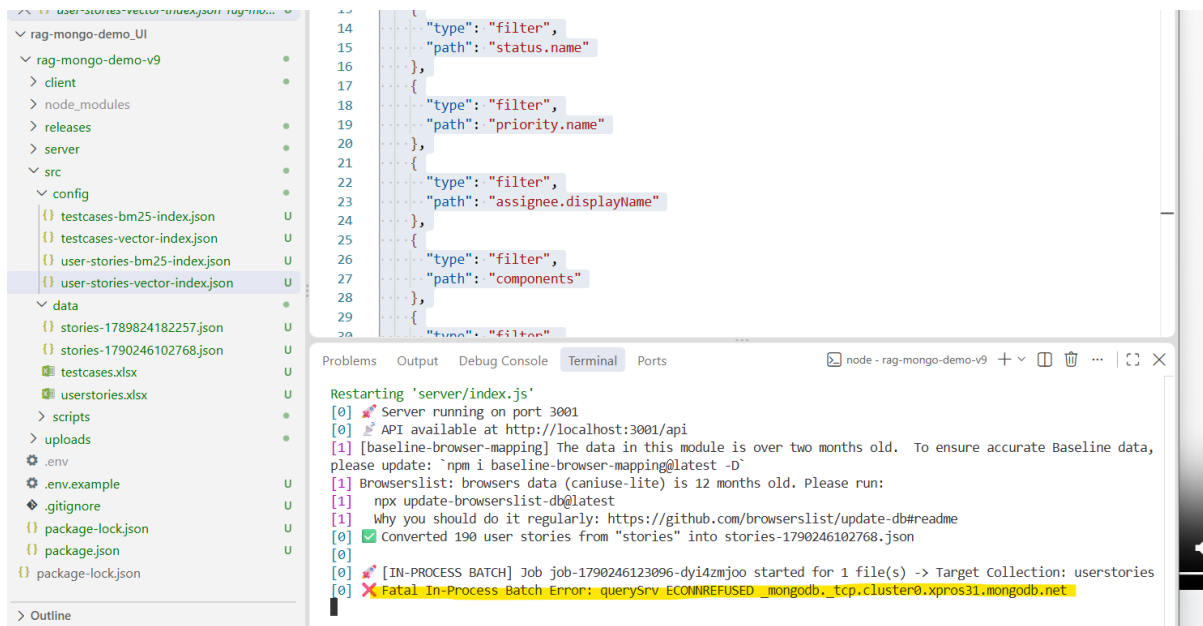
Short test: "Login test"

Detailed test: "Comprehensive user authentication test with validation steps"

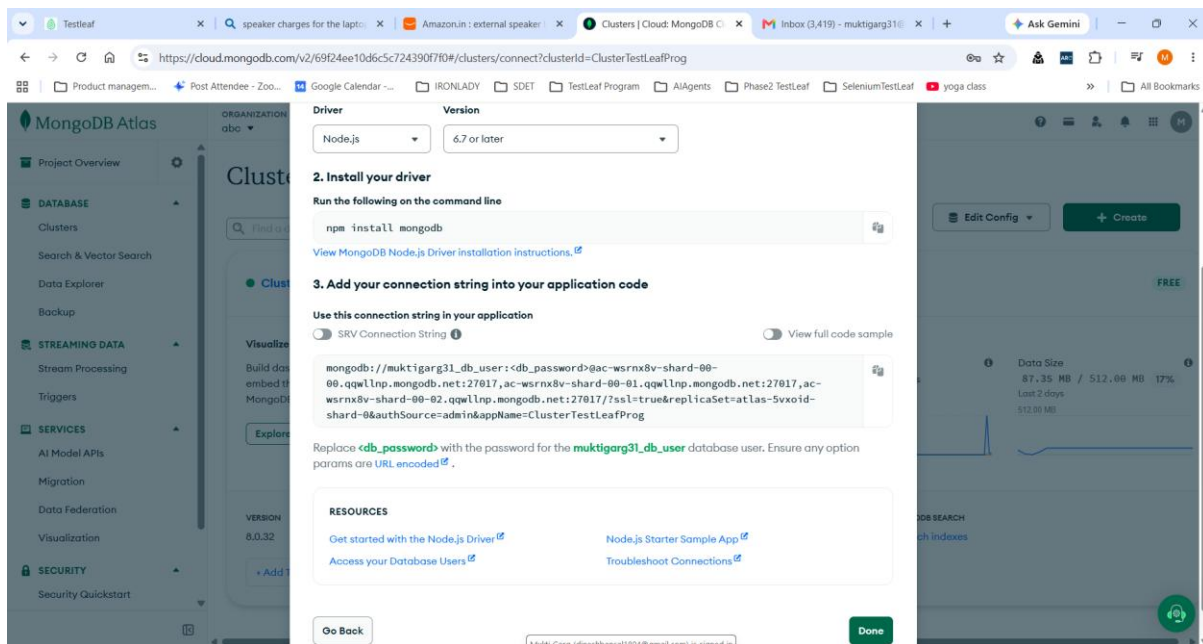
→ Different dot product scores (length matters)

✓ Best for: When you want both similarity AND completeness to matter

UI application



Updated the connection string



FileEditSelectionViewGoRunTerminalHelp

rag-mongo-demo-UI

rag-mongo-demo-v9

rag-mongo-demo-v9

package-lock.json

rag-mongo-demo-v9

1# ===== MongoDB Database & Collections =====
2# MongoDB Atlas Connection URI with credentials
3MONGODB_URI="mongodb://muktigarg31_db_user:0Kt3Xs0VpLKIHFXS@ac-wsrnx8v-shard-00-00.qqwillnp.mongodb.net:27017,ac-wsrnx8v-shard-00-01.qqwillnp.mongodb.net:27017,ac-wsrnx8v-shard-00-02.qqwillnp.mongodb.net:27017?ssl=true&replicaSet=atlas-shard-0&authSource=admin&retryWrites=true&retryReads=true"
4DB_NAME="RAG_testcases"
5# Database Name
6DB_NAME="RAG_testcases"
7
8# Test Cases Collection & Search Indexes
9COLLECTION_NAME="testcases"
10VECTOR_INDEX_NAME="vector_index"
11BM25_INDEX_NAME="bm25_search"
12
13# User Stories Collection & Search Indexes
14USER_STORIES_COLLECTION_NAME="userstories"
15USER_STORIES_VECTOR_INDEX_NAME="vector_index_userstory"
16USER_STORIES_BM25_INDEX_NAME="bm25_user_stories"
17
18# ===== Task-Specific LLM Models =====
19# 1. Primary Test Case Generation / QA LLM (groq | gemini | openai | claude | mistral)
20LLM_PROVIDER="groq"
21
22# =====
23

ProblemsOutputDebug ConsoleTerminalPorts

node - rag-mongo-demo-v9

[0] [IN-PROCESS BATCH] 190 user stories from "stories" into stories-1790246654440.json
[0]
[0] [IN-PROCESS BATCH] Job job-1790246668830-ivy9x0h0r started for 1 file(s) -> Target Collection: userstories
[0] Processing file (1/1): stories-1790246654440.json
[0] [Batch 1/4] Embedding 50 items with Mistral AI...
[0] [Batch 1/4] Inserted 50 documents.
[0] [Batch 2/4] Embedding 50 items with Mistral AI...
[0] [Batch 2/4] Inserted 50 documents.
[0] [Batch 3/4] Embedding 50 items with Mistral AI...
[0] [Batch 3/4] Inserted 50 documents.
[0] [Batch 4/4] Embedding 40 items with Mistral AI...
[0] [Batch 4/4] Inserted 40 documents.
[0]
[0] [IN-PROCESS BATCH COMPLETE] 190 items inserted, 0 failed, Total Cost: \$0.002417
[0]

Testleaf

speaker charges for the lapto...

Amazon.in : external speaker...

ClusterTestLeafProg Search |

Inbox (3,419) - muktigarg31@...

Ask Gemini

https://cloud.mongodb.com/v2/69f24ee10d6c5c724390f7f0#/clusters/atlasSearch/ClusterTestLeafProg

Product managem... Post Attendee - Zoo... Google Calendar... IRONLADY SDET Testleaf Program AI Agents Phase2 Testleaf SeleniumTestleaf yoga class

Search & Vector Sear...

ORGANIZATION abc PROJECT Project 0 CLUSTER ClusterTestLeafProg

Search Indexes

AUTO EMBEDDING

Usage

Rate Limits

NATIVE RERANKING

Usage

Rate Limits

SHORTCUTS

Cluster Metrics

Data Explorer

Search & Vector Search

Find a search index

VIEW COMPOUND QUERY EXAMPLE

CREATE SEARCH INDEX

M0 free clusters have a limit of 3 search and vector indexes per cluster. Indexes Used: 2 of 3.

Database	Collection	Index Name	Status	Queryable	Type	Index Fields	Documents (Estimated)
RAG_testcases	testcases	vector_index	READY	✓	vectorSearch	"embedding", "id", "module", "title", "description", "priority.name", "assignee.displayName", "assignee.name"	6,354 (100% indexed of 6,354)
RAG_testcases	userstories	vector_index_userstory	READY	✓	vectorSearch	"embedding", "jiraMetadata.projectKey", "status.name", "priority.name", "assignee.displayName", "assignee.name"	190 (100% indexed of 190)

Learn more about Atlas Search

Learn more about Atlas Vector Search

Alert System Status: All Good

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Testleaf x speaker charges for the lapto... Amazon.in : external speaker... ClusterTestLeafProg Data | Cl... Inbox (3,419) - muktigang31... Ask Gemini

https://cloud.mongodb.com/v2/69f24ee10d6c5c724390f7f0#/explorer/69f24fa90dcb47d5ea321867/RAG_testcases/userstories/find

Product managem... Post Attendee - Zoo... Google Calendar... IRONLADY SDET TestLeaf Program AIAgents Phase2 TestLeaf SeleniumTestLeaf yoga class All Bookmarks

ORGANIZATION abc PROJECT Project 0

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

ClusterTestLeafProg

- RAQ_testcases
 - testcases
 - userstories
 - admin
 - local

userstories

ClusterTestLeafProg > RAG_testcases > userstories

Documents 190 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA BULK EXPORT CODE

25 1 - 25 of 190

```
{ "_id": ObjectId("6ab4ff0efcf34fe8ab5970e0"),
  "key": "HC-257",
  "summary": "Consultant Dashboard My Inpatients",
  "description": "As a user, I want to consultant dashboard my inpatients so that the re...",
  "status": Object(2),
  "priority": Object(2),
  "assignee": null,
  "reporter": null,
  "created": "2022-01-06T00:00:00.000Z",
  "updated": "2026-09-24T10:44:14.586Z",
  "components": Array(empty),
  "labels": Array(empty),
  "fixVersions": Array(empty),
  "storyPoints": null,
  "project": "Health Care" }
```

Alert System Status: All Good

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